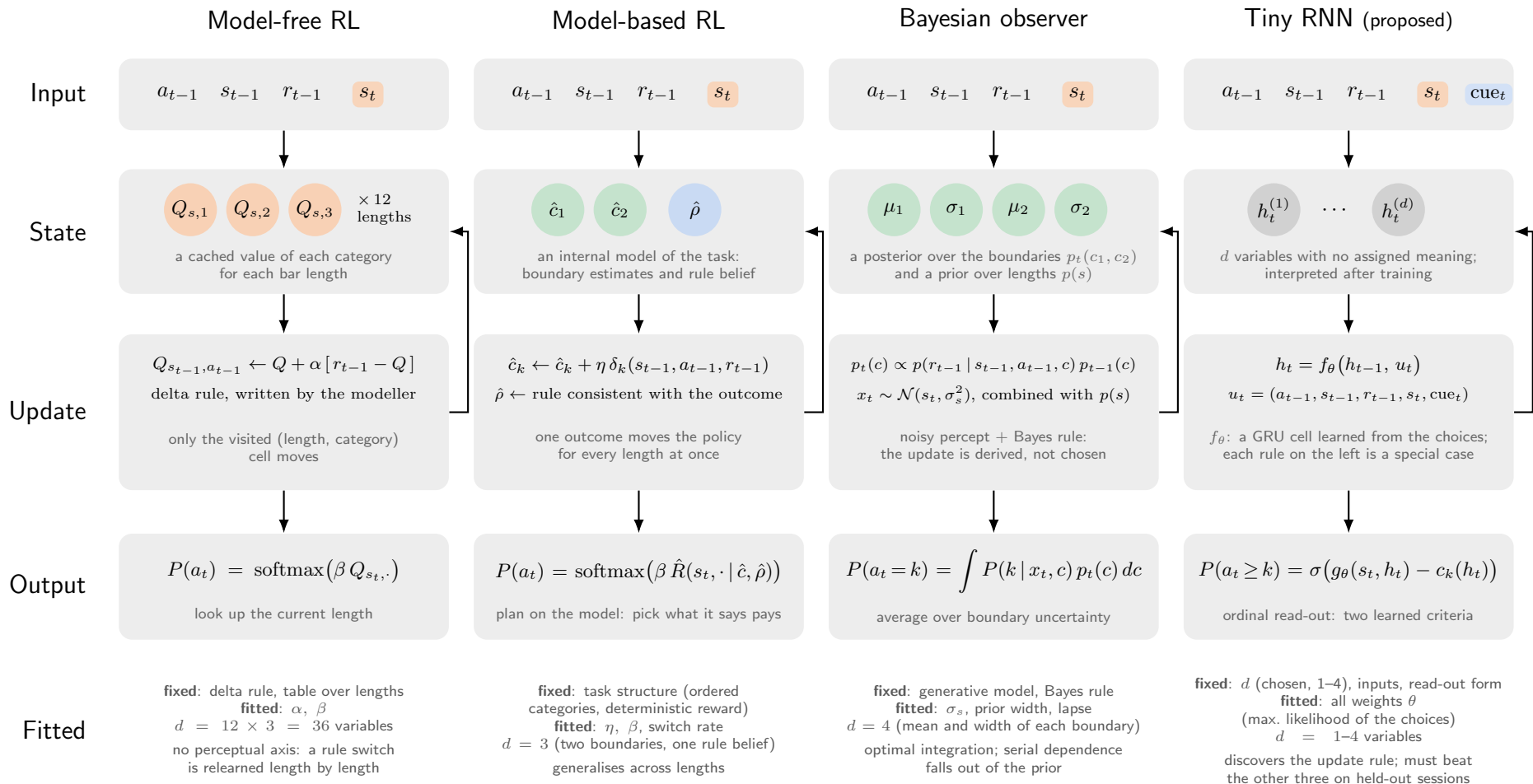


Same problem, four operationalisations: predict $P(a_t)$ from the trial history and the current bar



All four are dynamical systems on the same inputs with the same output. They differ in **what the state is**, **who writes the update rule** (the modeller, or the data) and **how many variables d** they carry. Following Ji-An et al., they are compared by held-out likelihood at matched d : a cognitive model that loses to an RNN with the same d has the wrong *form* of update, not too little memory.